

BiasAperture — Demographic Fairness Audit Dossier

Diagnostic Demographic Bias Evaluation & Regulatory Traceability Report

AUDITED MODEL
FairFace ResNet-34 Multi-Task Classifier

EVALUATION DATASET
FairFace Validation Benchmark (N=10954)

PROTECTED AXIS
Race

AUDIT DATE
2026-09-04 07:18:10 UTC

Diagnostic Scope & Non-Intervention Guarantee:

BiasAperture is engineered exclusively as an external diagnostic auditing, statistical significance testing, and regulatory traceability platform. It does **not** perform model retraining, in-processing debiasing, or synthetic image generation. In adherence to statistical integrity safeguards (NFR-003), demographic subgroups with fewer than 30 samples ($n < 30$) have point estimates suppressed to prevent unreliable compliance claims.

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SUBGROUPS WITH
SIGNIFICANT DISPARITY (P < 0.05)

Middle Eastern
LARGEST DEVIATION
SUBGROUP

DIR = 0.784
WORST HEADLINE METRIC

0
SUBGROUPS FLAGGED N < 30 (NFR-003)

1. Headline Fairness Metrics (Core Four — Cross-Group Overview)

Summary metrics evaluated across all eligible demographic strata ($n \geq 30$ per NFR-003). Accompanied by 95% stratified BCa bootstrap confidence intervals and Pearson's χ^2 test of independence. *Note: each CI band is scaled to its own row — dot colour (red = CI excludes null), not bar width, indicates significance across rows.*

Metric Name	Point Estimate	95% Bootstrap CI	CI Band vs. Null		Significance (p-value)	Regulatory Clause
Demographic Parity Difference	0.192	[0.156, 0.222]			p=0.0000 p < 0.05	EU AI Act Art. 10(2)(f) / NIST Measure 2.11
Equalized Odds Difference	0.053	[0.035, 0.068]			p=0.0000 p < 0.05	EU AI Act Art. 10(2)(g) / NIST Measure 2.11
Equal Opportunity Difference	0.053	[0.034, 0.069]			p=0.0000 p < 0.05	EU AI Act Art. 10(3) / NIST Measure 2.11
Disparate Impact Ratio	0.713	[0.676, 0.760]			p=0.0000 p < 0.05	EEOC 4/5ths Rule / EU AI Act Annex IV §2(g)

2. Per-Subgroup Disparity Matrix

Detailed stratification across individual demographic groups, ordered by largest deviation from the null (no-disparity) point, most significant first. Subgroups below the NFR-003 sample threshold ($n < 30$) are strictly flagged, uncomputed to prevent small-sample estimation artifacts, and summarised separately below.

Subgroup	Sample Size (n)	DPD	EOD (Max-Gap)	EOP	DIR (Symmetric)	Deviation from Null (red = $p < 0.05$)	Data Integrity Status
Middle Eastern	1209	0.145	0.016	0.010	0.784	DPD EOP EOD DIR	Audited (Valid)
Latino_Hispanic	1623	0.048	0.024	0.024	0.909	DPD EOP EOD DIR	Audited (Valid)
Indian	1516	0.034	0.012	0.004	0.936	DPD EOP EOD DIR	Audited (Valid)
Black	1556	0.026	0.034	0.034	0.951	DPD EOP EOD DIR	Audited (Valid)
White	2085	0.019	0.019	0.019	0.966	DPD EOP EOD DIR	Audited (Valid)
East Asian	1550	0.017	0.018	0.018	0.968	DPD EOP EOD DIR	Audited (Valid)
Southeast Asian	1415	0.013	0.009	0.000	0.975	DPD EOP EOD DIR	Audited (Valid)

3. Governance Documentation & Intended Use

EU AI Act (Regulation EU 2024/1689) Article 10 / Annex IV Alignment: This audit dossier provides verifiable diagnostic documentation of demographic performance disparities. It does not perform model retraining or in-situ weight modifications.

Model Description: ImageNet-pretrained ResNet-34 terminating in an 18-unit linear head sliced for race (7), gender (2), and age (9) classification.

Evaluation Protocol: Standalone offline evaluation using dual-backend cross-validation (Fairlearn + AIF360) and 1,000-resample stratified BCa bootstrap estimation.

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